

Overcoming the 3 dB squeezing extraction limit in silicon carbide microcombs with a photonic molecule: supplemental document

This supplemental document provides supporting content for the primary manuscript: the material dispersion models and their sources, the finite-element mode-solver settings with a mesh-convergence study, the dispersion-extraction procedure, the normalization and numerics of the modal Lugiato–Lefever solver including the co-moving-frame treatment of the drifting crystal, the field-normalization, Fourier, and detection conventions, the derivation and analytic validation of the linearized multimode quantum model, its relation to established multimode frameworks, the explicit two-ring model of the photonic molecule, the third-order-dispersion stability boundary, and numerical convergence studies for all quoted quantum quantities.

1. MATERIAL DISPERSION MODELS

Both material models come from the refractiveindex.info database, which is released to the public domain under CC0 [1]. The 4H-SiC ordinary-ray index uses the Sellmeier model of Wang *et al.* [2],

$$n_o^2(\lambda) = 1 + \frac{0.20075 \lambda^2}{\lambda^2 + 12.07224} + \frac{5.54861 \lambda^2}{\lambda^2 - 0.02641} + \frac{35.65066 \lambda^2}{\lambda^2 - 1268.24708}, \quad (\text{S1})$$

with λ in micrometres, valid from 0.4047 to 5 μm . The chip normal is the crystal c axis, so the fundamental TE mode is polarized in the ordinary plane and Eq. (S1) applies. The silica cladding uses the Malitson model [3],

$$n^2(\lambda) = 1 + \frac{0.6961663 \lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426 \lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794 \lambda^2}{\lambda^2 - 9.896161^2}. \quad (\text{S2})$$

At 1.55 μm these give $n_o(\text{SiC}) = 2.5644$ and $n(\text{SiO}_2) = 1.4440$. The Kerr index $n_2 = 6.9 \times 10^{-19} \text{ m}^2/\text{W}$ is taken from measurements on 4H-SiC [2, 4].

2. FEM MODE SOLVING AND CONVERGENCE

Modes are computed with the open-source solver `femwell` (v0.x, scikit-fem backend [5]) on triangular meshes generated by Gmsh [6]. The computational window spans $6.4 \times 4.9 \mu\text{m}^2$ around the core, with a core-region resolution of 35 nm for the geometry sweep and 40 nm for the selected-geometry runs, and graded coarsening into the cladding. Two modes are requested per solve and the TE-polarized mode is selected by its TE fraction, which exceeds 0.99 for all geometries.

Table S1 reports the mesh study at $\lambda = 1.55 \mu\text{m}$ for the selected geometry. With second-order elements the effective index is converged to below 10^{-7} between the two finest meshes, and the production setting (order 2, 40 nm) is indistinguishable from the finest run at that precision. The geometry sweep of the main text uses first-order elements for speed; the resulting absolute offset of about 6×10^{-4} in n_{eff} varies smoothly with wavelength and therefore has a negligible effect on the dispersion coefficients, which depend only on differences across the resonance grid.

3. DISPERSION EXTRACTION

The effective index is sampled at 17 wavelengths from 1.30 to 1.85 μm and fitted with a degree-six polynomial in a normalized frequency variable; the maximum fit residual is below 5×10^{-8} . Resonances of a ring of length $L = 2\pi R$ solve $\beta(\omega_m)L = 2\pi m$ with $\beta = n_{\text{eff}}\omega/c$, found by bracketed root finding on the fitted model for each azimuthal order. The radius is chosen so that the FSR at 1.55 μm equals 350 GHz, giving $R = 50.06 \mu\text{m}$. A quartic fit over $|\mu| \leq 40$ yields $D_2/2\pi = 6.89 \text{ MHz}$, $D_3/2\pi = -63.4 \text{ kHz}$, and $D_4/2\pi = -148 \text{ Hz}$. The LLE and quantum stages use the Taylor form of D_{int} , which agrees with the directly solved resonance grid to within the

Table S1. Mesh convergence of n_{eff} at 1.55 μm ($w = 1.85 \mu\text{m}$, $h = 500 \text{ nm}$, oxide clad).

core resolution (nm)	element order	elements	n_{eff}
60	1	3 426	2.3068171
50	1	3 426	2.3067888
40	1	4 612	2.3072447
30	1	6 326	2.3075482
50	2	3 426	2.3081254
40	2	4 612	2.3081259
30	2	6 326	2.3081259

fit residual over the simulated band. Bending corrections at $R \approx 50 \mu\text{m}$ are neglected; for this high-confinement core the bend-induced index shift is far smaller than the geometric dispersion tuning range explored in the design map.

4. LUGIATO–LEFEVER SOLVER

A. Normalization and mapping to dimensional amplitudes

Time is measured in units of $2/\kappa$, detunings as $\zeta_\mu = 2(\delta_0 + D_{\text{int}}(\mu))/\kappa$, and the field is scaled so that the self-phase-modulation rate equals $(\kappa/2)|\psi|^2$. In this convention the pump term is $f^2 = 8\eta g_0 P_{\text{in}}/(\kappa^2 \hbar \omega_0)$ with $\eta = 1/2$ at critical coupling. The single-soliton existence limit at fixed pump is $\zeta_0^{\text{max}} = \pi^2 f^2/8$, which for $f^2 = 9$ gives 11.1, in agreement with the observed annihilation point.

The solver stores modal amplitudes ψ_μ normalized to the N -point azimuthal grid; the physical modal amplitude used in Eq. (2) of the main text is

$$A_\mu = \sqrt{\frac{\kappa}{2g_0}} \phi_\mu, \quad \phi_\mu = \frac{\psi_\mu}{\sqrt{N}}, \quad (\text{S3})$$

so that $|A_\mu|^2$ is the intracavity photon number of comb line μ and every product $g_0 A_m A_n$ equals $(\kappa/2) \phi_m \phi_n$: in κ -normalized units, the couplings of the quantum model are $\frac{1}{2} P_{qk}$ and R_{qk} built from ϕ . Fluctuation operators obey $[\hat{b}_\mu, \hat{b}_{\mu'}^\dagger] = \delta_{\mu\mu'}$ and are Fourier transformed as $\hat{b}(\omega) = \int dt e^{i\omega t} \hat{b}(t)$. Quadratures are $\hat{x} = (\hat{b} + \hat{b}^\dagger)/\sqrt{2}$ and $\hat{p} = -i(\hat{b} - \hat{b}^\dagger)/\sqrt{2}$, and all quoted spectra are symmetrized two-sided spectra, averaged over $\pm\omega$ as measured by homodyne detection, normalized so that vacuum (shot noise) equals unity.

B. Numerics

Equation (1) of the main text is integrated with a symmetrized (Strang) split-step scheme: the linear part, including the pump, is advanced exactly in the modal basis, and the Kerr part is an exact phase rotation in the azimuthal basis. Production runs use $N = 192$ modes and $\Delta t = 0.002$. Table S2 shows that halving the step or increasing the grid changes the comb-line amplitudes by parts in 10^4 .

Table S2. LLE numerical convergence at $\zeta_0 = 6.5$ (modal amplitude of comb line $\mu = 2$, peak field, drift velocity).

N	Δt	$ \phi_2 $	$\max \psi $	v
192	0.004	0.342006	3.861400	-2.21707×10^{-2}
192	0.002	0.341966	3.861302	-2.21732×10^{-2}
192	0.001	0.341956	3.861196	-2.21738×10^{-2}
256	0.002	0.341966	3.841014	-2.21732×10^{-2}

C. Co-moving frame

With odd-order dispersion the crystal drifts at constant velocity v , so modal phases rotate as $\psi_\mu(t) = \psi_\mu(0)e^{-i\mu vt}$. We converge each state in a frame where a linear detuning term $-v\mu$ absorbs the drift, iterating between (i) relaxation under the modified detuning and (ii) measurement of the residual drift from the modal phase slopes. Three to four iterations reduce the stationarity residual $\max_\mu |\dot{\psi}_\mu|$ from order one to below 10^{-3} . The converged v equals the comb repetition-rate shift, $\delta f_{\text{rep}} = v(\kappa/2)/2\pi$, and vanishes to machine precision when D_3 is set to zero, an internal consistency check of the solver. States whose co-moving residual exceeds 10^{-2} are classified as breathing and excluded from the quantum analysis. The odd/even modal power contrast of converged crystals is about 220 dB, confirming exact C_2 symmetry.

5. LINEARIZED QUANTUM MODEL

A. Drift matrix

Substituting $\hat{a}_\mu = A_\mu + \hat{b}_\mu$ into the intracavity Kerr Hamiltonian and keeping quadratic terms yields Eq. (2) of the main text. In the doubled basis $\hat{\mathbf{v}} = (\hat{b}_{-M}, \dots, \hat{b}_M, \hat{b}_{-M}^\dagger, \dots, \hat{b}_M^\dagger)^\text{T}$ the dynamics is $\dot{\hat{\mathbf{v}}} = \mathbf{M}\hat{\mathbf{v}} + \Xi\hat{\mathbf{v}}_{\text{in}}$ with

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix}, \quad \mathbf{A}_{qk} = \delta_{qk} \left(-\frac{\kappa_q}{2} - i\Delta_q \right) + 2ig_0 R_{qk}, \quad \mathbf{B}_{qk} = ig_0 P_{qk}, \quad (\text{S4})$$

where Δ_q is the detuning of fluctuation mode q from the comb grid in the co-moving frame, $\Delta_q = (\kappa/2)(\zeta_q - vq)$ in normalized units. R is Hermitian and P symmetric by construction. For a 2-FSR crystal, P_{qk} and R_{qk} vanish unless q and k have equal parity, so the odd fluctuation sector decouples from the even one and carries no classical field.

B. Detection model

Each mode couples to an unmonitored intrinsic bath at rate κ_q^i and a monitored collection port at rate κ_q^c . Solving in the frequency domain gives the port scattering matrix, and the symmetrized quadrature spectral matrix of the detected modes follows from vacuum inputs on all ports, averaged over $\pm\omega$ sidebands as measured by homodyne detection. Squeezing values are eigenvalues of this matrix, normalized so vacuum equals unity; supermodes are its eigenvectors; and two-mode logarithmic negativities use the standard covariance formula [7]. In the adiabatic coupler model, odd modes take $\kappa^i = \kappa$ (all bare loss unmonitored) and $\kappa^c = \kappa_P$, after adiabatic elimination of a strongly damped auxiliary ring whose Purcell rate is $\kappa_P = 4J^2/\kappa_{\text{aux}}$; the explicit two-ring model of Section D removes this approximation.

C. Relation to established multimode frameworks

The supermodes quoted in this work are the eigenvectors of the real, symmetric quadrature spectral matrix $V(\omega)$ of the detected field. This is the standard frequency-resolved (morphing) supermode analysis of multimode quantum optics [8, 9]: at each analysis frequency, the eigenvectors are the joint quadratures a homodyne detector with a correspondingly shaped local oscillator would measure, and the extremal eigenvalues bound every measurable noise power. For a pure Gaussian output the same modes arise from the Bloch–Messiah reduction of the Bogoliubov (symplectic) transformation generated by the parametric interaction [10]; the two decompositions coincide because the covariance of a pure squeezed state is diagonal in its Bloch–Messiah basis. For the mixed state that reaches a real detector (unmonitored intrinsic loss, partial escape), the state-based eigendecomposition remains well defined and operational, which is why we adopt it. Loaded linewidths, hybridization, and mode-dependent decay enter $V(\omega)$ exactly, through the resolvent $(-i\omega - \mathbf{M})^{-1}$ in the input–output solution, so the frequency dependence of every supermode reflects the true modal response. As validation, the pipeline reproduces the analytic Bogoliubov eigenstructure of the cw-pumped resonator (Section E) and the two anchors of the founding soliton-crystal squeezing analysis [11] under matched assumptions: the 3 dB critically coupled bound and the growth of available squeezing with detuning.

D. Explicit two-ring model

For the full model of the photonic molecule, the doubled basis is extended by one auxiliary annihilation operator $\hat{a}_\mu$ per odd lattice site. The additional blocks of the drift matrix are diagonal,

$$\dot{\hat{a}}_\mu = -\left(\frac{\kappa_{\text{aux}}}{2} + i\Delta_\mu^{\text{aux}}\right)\hat{a}_\mu - iJ\hat{b}_\mu + \sqrt{\kappa_{\text{aux}}}\hat{a}_{\text{in},\mu}, \quad \dot{\hat{b}}_\mu = -iJ\hat{a}_\mu, \quad (\text{S5})$$

with the drop port monitored through κ_{aux} and the odd main-ring modes now fully unmonitored ($\kappa^i = \kappa$, $\kappa^c = 0$). The auxiliary detunings are $\Delta_\mu^{\text{aux}} = \Delta_\mu - \delta_\mu$, where $\delta_\mu = \omega_\mu^{\text{main}} - \omega_\mu^{\text{aux}}$ is the resonance mismatch computed by solving $\beta(\omega)L = 2\pi m$ for both rings (R and $R/2$) with the same FEM-fitted $\beta(\omega)$, after a single global thermo-optic offset chosen to minimize the worst-case $|\delta_\mu|$ over the analyzed lattice (Chebyshev alignment; the offset represents one microheater). A radius error δR scales the auxiliary FSR by $-\delta R/R_{\text{aux}}$, producing the linear walk-off of Fig. 6(a) of the main text. Vacuum inputs on all ports reproduce the identity output spectral matrix to 6×10^{-16} , and at fixed $\kappa_P = 4J^2/\kappa_{\text{aux}}$ the full model converges to the adiabatic result as $1/\kappa_{\text{aux}}$ (Table S3). Production numbers in the main text use $\kappa_{\text{aux}}/2\pi = 8$ GHz and $J/2\pi = 1.6$ GHz.

Table S3. Adiabatic-elimination convergence at $\zeta_0 = 6.5$, $\kappa_P = 10\kappa$: peak drop-port squeezing of the full two-ring model versus auxiliary linewidth (perfect alignment). The adiabatic value is -8.49 dB.

$\kappa_{\text{aux}}/2\pi$ (GHz)	2	4	8	16	64
peak squeezing (dB)	-6.84	-7.35	-7.89	-8.18	-8.41
deviation (dB)	1.65	1.14	0.60	0.31	0.08

E. Analytic validation

The implementation passes four checks. (i) With no comb the output spectral matrix equals the identity at all frequencies to machine precision ($< 3 \times 10^{-16}$). (ii) For a cw-pumped resonator the drift eigenvalues of the $(\mu, -\mu)$ pair reproduce the analytic Bogoliubov result $\lambda = -\kappa/2 \pm \sqrt{g^2 - \delta^2}$ with g the parametric gain and δ the shifted detuning, to 10^{-14} . (iii) At critical coupling the computed output squeezing never exceeds the analytic 3 dB bound, the Heisenberg product $\min V \times \max V \geq 1$ holds on all frequency grids, and over-coupling raises the detected squeezing as expected. (iv) With two equivalent frequency-flat baths of which one is monitored, the detected and perfectly monitored spectral matrices obey the noise-flow identity of Eq. (3) of the main text, $V_{\text{out}}(\omega) = (1 - \eta)\mathbb{1} + \eta V_{\text{ideal}}(\omega)$; the solver satisfies it to better than 10^{-8} per frequency and quadrature, for the bare device ($\eta = 1/2$) and the adiabatic molecule ($\eta = 10/11$) alike.

F. Truncation study

Production runs keep fluctuation modes $|\mu| \leq 30$ and classical comb modes $|\mu| \leq 60$. Table S4 shows the dependence of the peak squeezing on the truncation M . The bare-device value oscillates by about 0.1 dB, limited by the edge of the truncated lattice; the molecule value converges from below with a change of 0.22 dB from $M = 30$ to $M = 40$. We therefore quote molecule squeezing with a numerical uncertainty of about 0.2 dB; the production values are slightly conservative.

Table S4. Peak squeezing versus fluctuation-mode truncation at $\zeta_0 = 6.5$.

M	bare (dB)	molecule $\kappa_P = 20\kappa$ (dB)
20	-2.886	-6.616
30	-2.751	-7.275
40	-2.963	-7.498

6. THIRD-ORDER-DISPERSION STABILITY BOUNDARY

The D_3 robustness study of the main text (Fig. 7) scales the FEM value $D_3/2\pi = -63.4$ kHz by factors 0, 1, 2, 2.5, and 3 at $\zeta_0 = 6.5$. Table S5 lists the exact peak drop-port squeezing for each

scale (adiabatic model, $\kappa_P = 20\kappa$); the full spread is below 0.01 dB. The stability boundary is located by continuing the scan in steps of 0.25 in the scale factor: the converged crystal survives at $3.25\times$ (co-moving residual 6×10^{-4}) and fails the stationarity criterion at $3.5\times$ (residual 0.22). The boundary is unchanged on azimuthal grids of $N = 192, 256$, and 320 modes, which excludes a truncation artifact.

The loss mechanism is a dispersive-wave resonance. The phase-matching mode where $D_{\text{int}}(\mu) = 0$ (including D_4) moves from $\mu = 98$ at $3.25\times$ to $\mu = 91$ at $3.5\times$, entering the crystal's spectral support. Seeding the converged $3.25\times$ crystal at $3.5\times$, the state develops a strong comb-line shoulder near the phase-matched mode and settles into a breathing state: the peak field oscillates by about $\pm 17\%$ at 58 MHz while the odd/even modal contrast stays below -260 dB, so the state remains an exact C_2 -symmetric crystal that has undergone a Hopf bifurcation rather than a parity-broken state. Parity itself breaks by $4\times$ the design value, where the odd-mode contrast collapses to about -21 dB.

Table S5. Peak drop-port squeezing versus D_3 scaling (adiabatic model, $\kappa_P = 20\kappa$, $\zeta_0 = 6.5$), as plotted in the inset of Fig. 7(b) of the main text.

D_3 scale	0	1	2	2.5	3
peak squeezing (dB)	-7.274	-7.275	-7.278	-7.280	-7.284

7. SUPPLEMENTARY FIGURE

Figure S1 shows the supermode structure at the molecule drop port and a direct comparison of bare and extracted squeezing spectra. The two-supermode structure of the bare device survives Purcell extraction, and the spectra broaden as the odd-mode linewidths grow.

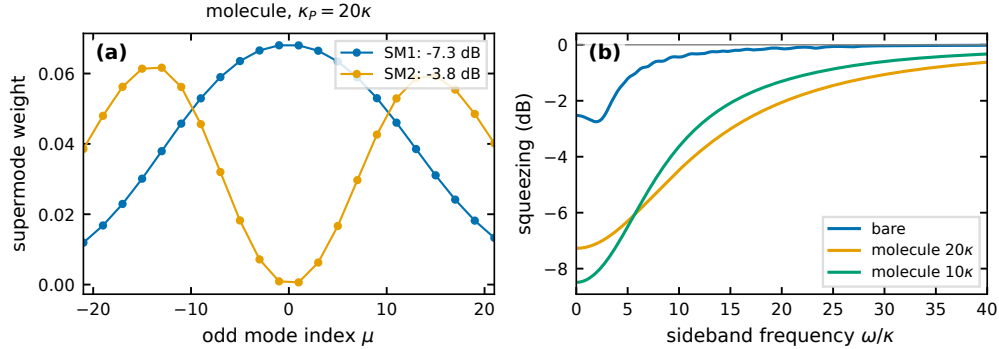


Fig. S1. (a) Weights of the two dominant squeezed supermodes at the drop port for $\kappa_P = 20\kappa$. (b) Optimal squeezing spectra of the bare device and the molecule at $\kappa_P = 10\kappa$ and 20κ ($\zeta_0 = 6.5$, pump 8.3 mW).

8. REPRODUCING THE RESULTS

The simulation code is openly available on GitHub at github.com/Tanvir-Mahmud-Mahim/sic-molecule-squeezer (Apache-2.0 license), and the complete raw data together with the validation testbench and its reference outputs are archived on Zenodo at [doi:10.5281/zenodo.21471674](https://doi.org/10.5281/zenodo.21471674) (CC-BY-4.0). The released package contains the complete pipeline: `materials.py` (Sellmeier models), `fem_modes.py` and `sweep_fem.py` (FEM), `dispersion.py` (resonance grid), `lle.py` and `exp_lle.py` (mean field), `exp_d3boundary.py` (D_3 stability boundary), `quantum.py`, `exp_quantum.py`, and `exp_addendum.py` (quantum stage), `aux_ring.py` and `exp_tworring.py` (two-ring alignment and full molecule model), `validate_quantum.py` and `convergence_checks.py` (verification), and `make_numbers.py`, which regenerates every number quoted in the manuscript directly from the simulation outputs. The testbench pass criteria are: vacuum output equal to the identity to better than 10^{-15} ; Bogoliubov drift eigenvalues matching

the closed form to 10^{-13} ; critically coupled output squeezing never exceeding the 3.01 dB bound; Heisenberg product $\min V \times \max V \geq 1$ on all frequency grids; and the convergence tolerances of Tables S2 and S4.

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